

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTES305 - ENGINEERING MATERIAL SCIENCE

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The electrical conductivity of a metal decreases with:
A) Decreasing temperature B) Increasing temperature
C) Increasing pressure D) Decreasing impurity
- Dielectric constant is defined as:
A) Ratio of magnetic flux to field
B) Ratio of capacitance with dielectric to without dielectric
C) Conductivity of material
D) None of the above
- 3 In intrinsic semiconductors, the number of electrons is:
A) Greater than holes B) Less than holes
C) Equal to holes D) Zero
- 4 Magnetic hysteresis loop gives information about:
A) Conductivity B) Magnetization behavior
C) Polarization D) Ionization energy
- Galvanization is done to prevent:
A) Electrical conduction B) Rusting
C) Radiation D) Heat loss
- 6 Structural materials are selected based on:
A) Only color
B) Cost only
C) Mechanical strength and durability
D) None of the above
- 7 Magnetic tapes use:
A) Soft magnets B) Hard magnets
C) Non-magnetic materials D) Semiconductors
- 8 High-speed steel is an example of:
A) Ferrous alloy B) Non-ferrous alloy
C) Semiconductor D) Dielectric

- 9 The Fermi level in an n-type semiconductor lies:
 A) Near valence band B) At mid-gap
 C) Near conduction band D) Outside energy bands
- 10 Piezoelectric materials generate:
 A) Heat on pressure B) Electric charge on pressure
 C) Light on current D) Pressure on magnetism
- 11 Superconductivity is observed when:
 A) Temperature is high
 B) Impurities are high
 C) Temperature is near absolute zero
 D) Magnetic field is increased
- The mean free path of an electron is the:
 A) Distance between two collisions
 B) Average energy of electrons
 C) Velocity of electrons
 D) Drift velocity

Q2. Solve the following.

- A) Explain the effect of temperature and impurities on the conductivity of metals 6
 B) What is dielectric loss? Discuss factors affecting dielectric strength. 6

Q3. Solve the following.

- A) Write short notes on:
 a) Joule's Law b) Superconductivity and its applications 6
 B) Explain various types of polarization in dielectric materials with examples. 6

Q4. Solve Any Two of the following.

- A) Write a note on internal fields in solids and their effects on dielectric behavior. 6
 B) Explain the mechanism of conduction in semiconductors with energy band diagrams. 6
 C) Differentiate between soft and hard magnetic materials with examples. 6

Q5. Solve Any Two of the following.

- A) What is magnetostriction? Mention its applications. 6
 B) What are refractory materials? Explain their types and applications. 6
 C) A magnetic material is subjected to a magnetizing field of $H=1000 \text{ A/m}$ and produces a flux density of $B=1.5 \text{ Tesla}$. Calculate the magnetic permeability μ and relative permeability μ_r 6

Q6. Solve Any Two of the following.

- A) Explain the aging of permanent magnets and methods to reduce it. 6
 B) Explain the process and advantages of impregnation of materials. 6
 C) Write notes on diamagnetic and paramagnetic materials with atomic interpretation. 6

*** End ***